Interactive speech systems for telecommunications applications

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Speech technology provides a powerful tool for the development of new products and services, particularly within the telecommunications sector. This paper considers current speech technology and discusses approaches to the design of successful interactive speech systems. A number of examples are given of existing services and advanced prototype systems.

1. Introduction

The convergence of telecommunications and computing in the latter half of this century has probably had a greater impact upon society than almost any other technological change within that period. It has lead to an explosive growth in the availability and use of telecommunications and unprecedented demands and opportunities for sophisticated new products and services. Communication has become a global business, with the potential to involve almost everybody on the planet. This growth offers both challenges and opportunities, as a broadening base of users from diverse backgrounds demand increasingly varied products and services of a high quality, at a reasonable price and which are straightforward to use, and as this expansion drives a need for increased automation.

Speech interfaces build upon a basic human skill. The evolution of spoken language has given us an effective mechanism both for giving instructions and for expressing and interpreting ideas. Therefore speech technology can provide a flexible and intuitive approach to the control of advanced services and interface design can help to produce products and services which are straightforward to use and require little user training.

As developments in processor and storage technologies continue to drive down the cost of computing power in the network, on the desktop and in consumer and telecommunications equipment, speech will increasingly become the primary mode in many user interfaces across a wide range of market segments:

- in the office, voice control of personal computers and voice dictation is set to become more widespread,
- in the home, speech may become commonplace for the control of multimedia systems such as advanced interactive television,
- on the move, the convergence between mobile telephones and personal digital assistants (PDAs) will drive the development of ubiquitous voice and graphics driven communication assistants.

However, perhaps most dramatically, continued and rapidly accelerating growth will be seen in the use of speech technology within and at the periphery of the telecommunications network. Speech provides a method by which nearly everyone can control an increasingly feature-rich network, access sources of information and interact with customers and suppliers. Interactive voice response (IVR) systems can enable companies to re-engineer the way in which they do business with their customers, and it provides an approach to handling the explosive growth in the telecommunications and information sectors [1, 2]. It is likely that, despite the rapid growth in on-line access, in the medium term the majority of enquiries and transactions will occur by phone and much of it will rely on IVR technology. For example it has been estimated that within the US, 70% of call-centres employ IVR systems.

This paper concentrates on the development of speech applications within the telecommunications area. It discusses relevant aspects of the technology and issues of integration and service design.

2. Telecommunications applications of speech technology

Speech technology is particularly relevant within the current telephone network as it can be used to:

- provide an intuitive interface,

- provide access to services without the need for additional customer equipment, with relevance both to customer acceptability and mobility,
- replace or supplement services which currently require the use of human operators,
- reduce the cost of service provision — new services can be inexpensively developed and made available through the international network.

Across a broad range of businesses, including telcos, the primary drivers for the introduction of automated services are:

- cost reduction,
- new revenue opportunities,
- increased customer quality and flexibility such as 24-hour operation.

Specifically within operator and call-centre applications, automation through the use of speech technology can allow people to handle individual calls more efficiently and be freed from many routine tasks to concentrate on high-value activities or problematic calls. This in turn allows flexibility in handling market growth and the ability to respond rapidly to new market opportunities. Similarly, new information services can be developed and deployed with reduced requirements for costly operator infrastructure, and existing manual services can be expanded to take advantage of prevailing market conditions or new customer groups.

IVR technology can enable a wide variety of applications, for example:

- advanced call handling facilities

  — voice controlled dialling,

  — auto-attendant/call routeing,

  — enhanced network services;
- messaging

  — voice-mail,

  — voice access to electronic mail,

  — network message delivery;
- information and entertainment

  timetables,

  directories,

  music browsing and ordering;
- transaction processing

  — home banking/home shopping,

  — work-force management,

  — automation and enhancement of operator services,

  — call-centre automation,

  — advertising tie-ins,

  — ordering of entertainment services.

IVR systems may be categorised by the primary mode of user input, typically either TouchTone ® or speech recognition. The following sections describe their use in more detail.

2.1 TouchTone services

For some years TouchTone has been used as the primary method for controlling network-based speech applications, providing inexpensive solutions to a wide variety of applications. Often, a menu structure is implemented and the caller is guided through the structure with prompts of the form:

> ***For News, press 1,***
> ***sport 2,***

TouchTone can provide a good interface for motivated users and conventions for best practice in the design of dialogues have been established. Standards are available for the assignment of keys to commonly used functions and are discussed in section 3.3.1. However, the mapping of system functions on to digits is arbitrary and not necessarily intuitive for complex tasks.

For many applications, TouchTone provides an appropriate and effective form of input. For example, the entry of a security PIN is well known to a public used to automated telling machines, and may give an increased sense of security to some users. Where long digit strings must be entered, for example card or telephone numbers, the greater speed and accuracy of TouchTone, may mean that it is preferred.

TouchTone can be used for the entry of alphabetic information, for example entering surnames to select a mailbox on a corporate Voice-Mail server. However, within the UK, few phones are marked with the letters and internationally not all labelling is consistent. Some applications rely upon more than one key press for each letter, which can be difficult and cumbersome for callers.

In some countries, penetration of TouchTone-capable telephones and even the proportion of capable telephones which are not correctly configured can restrict the availability of TouchTone-based services. Other issues such as the unavailability of the '*' key in some countries may need to be taken into account.

2.2 Speech recognition for telephony services

For some applications, speech recognition can be used to provide a more user friendly interface. For example, as discussed above, once alphanumeric or large vocabulary input is required, TouchTone becomes less appropriate.

In many cases, system designers may decide to offer speech and TouchTone in parallel, allowing the user to select the method with which they feel most comfortable. This may or may not be explicit in the outgoing prompts. In general, simply overlaying a dialogue based upon TouchTone menus with spoken digits is unlikely to result in an improved interface. For example:

> ***For News, say or press 1,***
> ***Sport 2,***

If possible, the natural vocabulary of the application should be reflected in the spoken vocabulary of the system. It is now possible to rapidly generate arbitrary recognition vocabularies which are intuitive within the particular application and to allow the entry of large-vocabulary information such as names. Where possible, alternative responses which may be reasonably expected from users should be catered for, e.g. 'football' and 'soccer'.

> ***Please say the name of the service you require:***
> ***news, weather or sport***

Speech recognition may be preferred for hands-free and eyes-free applications, such as those within cars. Also for use by cellular phone users, when buttons are awkwardly positioned for use in interactive systems.

3. Speech technology

This section discusses the main components of IVR systems which use speech recognition. TouchTone systems typically contain many of the same basic components. Interactive speech systems comprise:

- dialogue controller — responsible for controlling the interaction with the caller, for example, maintaining the dialogue logic, sequencing prompts, selecting the recognition vocabulary, interpreting the semantics of user responses and determining the appropriate system operation,
- speech output — responsible for output messages and prompts,
- speech input — responsible for interpreting the callers responses, often supplemented with TouchTone input for digit string entry and with 'speaker verification' for enhanced security,
- information manager — interfaces with the host application or application database and acts as a mediator for high-level dialogue requests, for example, building recognition vocabularies based upon database contents.

The separation between dialogue and information management may be trivial in a simple application and both activities may be handled by a single agent. This issue is discussed in more detail for advanced applications, such as directory searching, in Attwater and Whittaker [3]. Figure 1 shows this diagrammatically, and the following sections discuss the speech and dialogue components separately and then issues of integration.

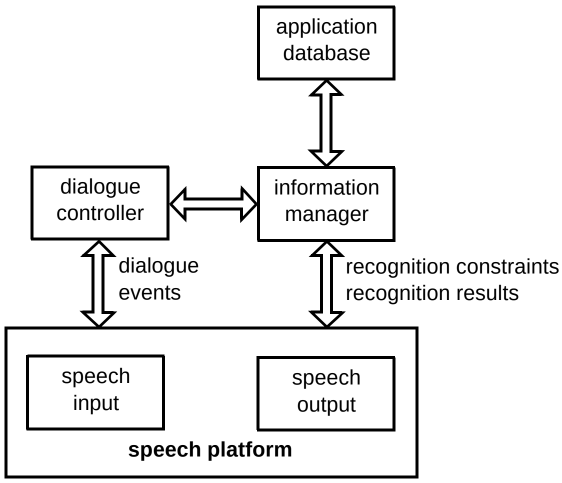

Fig 1 Key components of speech applications.

3.1 Speech input technology

Most speech recognition systems currently in use are based upon statistical classification techniques such as hidden Markov modelling (HMM) [4]. This is based upon building mathematical models from an explicit training phase. At run time, the task of the recogniser is to determine which model or sequence of models as defined by some lexicon or grammar (finite state or statistical) best explains the observed speech.

The performance of such systems depends upon how representative the training data and grammars are of the utterances which will be encountered in live use. Ideally, therefore, the initial training set should be as representative as possible of speech from the types of applications for which the recogniser is going to be used, for example telephony versus microphone speech, isolated word versus continuous speech.

The effective use of speech recognition is based upon the ability to predict and model the responses which are expected from the user. Hence the performance of an application which uses speech recognition is highly dependent upon the ability of the dialogue to steer the user to give responses in a predictable fashion.

Speech recognition is not a precise process. A certain error rate must be expected and should be catered for in the user interface design. This is particularly the case in telecommunications applications where the user population, background noise conditions, telephones and channels can vary widely and be difficult to predict. The dialogue should generally be designed to identify and repair such errors through the use of recognition confidence measures, confirmation and re-prompting.

Speech recognition systems can be categorised in a number of ways, and in general, increased functionality results in higher computational and storage requirements and hence, in the network context, per-line costs.

3.1.1 Types of recogniser

Speaker-dependent recognisers are trained specifically to the characteristics of a particular speaker. This involves collecting training data for that speaker. This technology has been largely superseded for telephony applications, with the notable exception of speaker adaptation described below.

Speaker-independent recognisers are trained on a corpus of data collected from a variety of speakers and can be used by any previously unseen speaker (given that the training speakers are representative of those expected in the live application). Speaker-independent systems form the basis of the majority of new services.

Speaker-adaptive recognisers adapt the characteristics of a speaker-independent recogniser as more information becomes known about the user. This technology has been used in systems where a single user uses the system extensively, e.g. voice dictation, although it is not yet commonplace in telephony applications.

3.1.2 Fluency

Systems can be designed which guide users to speak in different styles, for example as isolated words or in fluent speech. Different recognition techniques and components are required dependent upon the fluency of the speech expected and the detail of the interpretation which is required.

- Grunt detection

  The recogniser only identifies that something has been spoken. It does not attempt to identify the sound as a word or utterance. This can be used to interrupt long messages or to indicate a desired option from a menu.

- Isolated-word entry

  A single word or well-defined phrase is spoken in response to a prompt.

- Word spotting

  A single word or well-defined phrase is identified in a string of words.

- Fluent-string entry

  A sequence of letters or numbers is spoken in a stream. This is useful for entering spellings and alphanumeric account numbers, PINS and telephone numbers.

- Fluent speech

  Complete phrases or sentences may be spoken.

At present most applications are implemented using isolated-word entry. Although fluent-string entry is increasingly used, fluent speech systems are only now becoming a realistic proposition for use in network applications within tightly restricted domains.

Fluent speech places very different requirements on other components of the system particularly the dialogue controller. For example, more than one piece of information can be given in a single response, or the user may refer to a piece of information given in a previous response. In addition to the basic recogniser, a parser is required to both identify words and extract the meaning from the sentence or phrase recognised.

In reality, these modes form a continuum. For example, if an isolated-word recogniser is to be robust to the variety of responses which users will actually give in a live application, it has to embody some degree of word spotting just to isolate the word from the surrounding noise and carrier phrases. It can be more helpful to consider the mode of entry as being more an issue of dialogue design, comprising the estimation of predicted responses, than one of recognition technology.

Good dialogue design can ensure that the majority of users can be primed and guided to respond in a constrained fashion. However, a proportion will always respond in unexpected ways, and it is therefore important that the recogniser is robust and can either ignore extraneous portions of a response or can detect and reject unexpected responses. It can be as important in an application to know if a response has been effectively recognised, as to know what the interpretation was. Dialogue decisions such as re-prompting can be made dependent upon such confidence criteria.

3.1.3 Vocabulary creation

Within an application, the vocabulary which will be used by the caller must be specified and used to configure the recognisers. This can be done in one of two ways:

- **whole-word** recognition training relies on the collection of multiple (tens or hundreds) examples of each word within the application,

- **sub-word** recognition training is based upon the construction of new vocabulary words from a pretrained vocabulary-independent set of phonetic (or other) sub-word units, using a pronunciation dictionary or lexicon.

Except where large pre-existing databases exist for a particular vocabulary as in the case of digits and 'yes/no', sub-word training based on an application-independent training database is now the preferred approach for the development of new applications. This is because the approach allows vocabularies to be defined and recognisers to be configured either by the designer or by reference to some application database, without the need for a specific speech data collection. It is therefore suitable for the construction of large vocabularies such as those required for directory access, catalogue shopping, etc.

An approach referred to as 'rapid vocabulary generation' allows new recognition vocabularies to be specified by the dialogue designer with little expert knowledge of the speech recognition component, simply by typing in the words of the vocabulary.

In practice, application-independent training databases may not optimally represent either the vocabulary of a particular application, or the way in which users respond. Therefore, once an application has been running for some time, application-specific training data should be collected and fed back into a retraining cycle, building either improved (application-specific) sub-word or whole-word models.

Sub-word recognition relies on the availability of high-quality phonetic lexicons which specify the pronunciations associated with potential vocabulary items in terms of phoneme sequences. These must capture both regional and international variations and the variations in pronunciation which may be reasonably expected within the application. This is particularly the case with real names, which show wide variations. Pronunciations can be generated in a number of ways:

- 'hand-crafting' based upon linguistic expertise,

- automatically by rule,

- automatically by analogy with similar vocabularies,

- automatically by auto-transcription of spoken exemplars using a speech recogniser.

Typically, the vocabulary of an application can be specified as a part of the application and dialogue design. There are, however, other instances where this is not the case and alternative approaches have to be taken.

- User-selectable vocabulary

  This is the situation in which a user can specify their own vocabulary items for use in, for example, voice-controlled dialling applications (e.g. 'call Bob'). Typically, this requires the user to repeat the word or phrase a small number of times, under the guidance of the dialogue. Alternative approaches to implementation allow the resultant vocabulary to be speaker-dependent or speaker-independent.

- Database access

  In applications where the vocabulary is changing rapidly, such as directory or customer identification applications, the system must be able to dynamically update its active vocabularies. Database applications are discussed in Attwater and Whittaker [3].

3.1.4 Vocabulary size

The size of vocabulary which can be supported within an application depends upon factors including:

- channel and terminal characteristics,

- confusability of vocabulary,

- available computing power,

- quality, quantity and applicability of training data,

- user population.

Although vocabulary requirements will vary from application to application, the data in Table 1 is indicative of the size of vocabularies which are currently available for speaker-independent telephony recognition supported on platforms such as the interactive speech application platform [5, 6] (ISAP).

Table 1 Example vocabulary sizes.

Mode	Vocabulary size
isolated yes/no	about five
isolated digits	twelve
isolated command words	tens
large vocabulary (e.g. names)	hundreds — thousands
fluent spelling and alphanumeric strings	tens of thousands of words
fluent digit strings	unconstrained

In general, increasing the size of a recognition vocabulary will reduce the accuracy of the recogniser. The similarity of the words also has a significant effect.

Attwater and Whittaker [3] discuss approaches which can be taken to combine the results of a number of different large vocabulary recognisers, arbitrarily spoken or spelt, to raise system performance in applications where each individual vocabulary is too large to achieve adequate accuracy in isolation.

Alternatively, recognisers can be configured to focus on the most common subset of responses expected at a particular point in a dialogue and to exhibit aggressive rejection or utterances outside of that subset. Such ‘partial vocabulary recognisers’ can be employed in applications where a small subset dominates what is, overall, a large vocabulary.

3.1.5 Speaker recognition

Speaker recognition is the term used to refer to both speaker verification and speaker identification. The technology is closely related to speech recognition.

Speaker verification is the process of validating or rejecting that a person is who they claim to be. For example, in a financial or calling-card application, where a caller has attempted to identify themselves by the entry of a card or account number, that claim can be validated using speaker verification. Speaker verification can have an impact on the perception of the level of security of a system. In some cases, TouchTone entry of a security PIN may be inappropriate (for example some hotel PBXs can record a calling-card PIN and display it on the bill) and so spoken security may be preferred. In this case, speaker verification adds an additional layer of security which can detect if an impostor has obtained the PIN.

Speaker identification is the process of identifying an individual from a set of speakers without any prior identity claim. This can be used, for example, to identify a family or team member using a shared system.

Just as applications using speech recognition have to cope with a certain error rate, so speaker verification has a corresponding error rate in terms of incorrect rejections of the real user and incorrect acceptances of impostors, and speaker identification will have a certain level of false categorisation.

Typically, speaker-recognition systems require a separate enrolment dialogue in which the characteristics of the caller are obtained. In general, the performance of both technologies is dependent upon the amount of training material that is available during enrolment and on the amount of information available in the verification/identification pass. A balance has to be struck, dependent upon the particular application, between the amount of intrusion which the procedure causes within the dialogue and the performance required. In some cases, it may be possible to perform recognition and verification of the same utterance, for example, when an identification number is being entered.

Speaker recognition is not yet as mature a technology as speech recognition, and is only beginning to ease itself into applications. It is worth bearing in mind the difficulty of this task for people and how easily they can be fooled.

3.2 Speech output technology

Traditionally most IVR systems have used pre-recorded messages for dialogue prompts and information recordings. Often professional speakers are used to give a high quality of output and the same speaker or set of speakers may be used for a number of applications both to maintain a corporate identity and to ease the transition between services where one service gateways into another. In some cases, more than one speaker may be used within a single application to distinguish different activities or phases. For example, help messages may be spoken by a different voice with the purpose of indicating the structure of the dialogue to the caller.

As well as providing prompts, pre-recorded digits, words and phrases can be concatenated to enable the output of information such as telephone numbers as in the automatic announcement subsystem used within the BT directory assistance system. Care is needed in the selection of component phrases to ensure that the resultant concatenation is of a suitably high quality. For example, in the case of times or phone numbers, each word or number may be recorded in a number of contexts and the most appropriate unit selected at run time.

In situations where it is impractical to use concatenated recordings, a more flexible output system is required, capable of dealing with arbitrary output phrases. Such applications are those where:

- the information is changing rapidly,
- the information is dynamic and varied across users, such as reading out electronic mail,
- the vocabulary of the system is just too large to be practically recorded or stored such as directory applications.

Text-to-speech (TTS) synthesisers which are capable of speaking arbitrary sentences have been available for some years, but until quite recently, the quality has been very limited. While the available quality has been acceptable for high-value applications, such as those for the disabled or closed user group financial services, it has not been appropriate for the majority of telephony applications. However, the new generation of synthesisers, such as BT's Laureate system have achieved a much higher level of performance and may be considered for use in advanced applications.

The combination of high-quality recorded messages with TTS generated speech in another voice should be used with care. However, BT's Laureate TTS system can be trained to sound like a particular individual, which allows the combination of recorded messages from a given speaker with inserted segments generated in the same voice to cater for variables within an output phrase. For example:

'The number for' <John Smith> 'is' <01473 ...>

where recorded messages are shown quoted and TTS variables bracketed.

3.3 Dialogue management

3.3.1 Dialogue design

Dialogue design is often the key to successful speech system design. In any IVR application it is the core of the system and specifies the user interface which the caller will encounter.

A significant amount of work has been undertaken at BT on the development of best-practice guidelines for the design and implementation of TouchTone and speech dialogues. Internally, BT designers have access to the BT voice application style guide which is based on extensive research and field experience and a number of external guides [7] have been produced to assist designers. In addition, BT is a partner in the Dialogues 2000 project [8] and funds a significant amount of research work at the Centre for Communication Interface Research at the University of Edinburgh [9]. A number of other standards projects, such as the EU-funded EAGLES project, are also under way. However, only a small number of international standards exist, such as ETSI/ISO standards for key assignment of standard functions within TouchTone dialogues, for example '1' for yes and '2' for no.

Generally, a successful dialogue will be one in which the user feels:

- comfortable,
- that they are in control, but being guided where necessary,
- that they know how to respond to any question,
- that they can get help at any point,
- that they can exit conveniently.

In brief, it must be clear and consistent.

The approach taken to the design of the structure of a dialogue will depend upon the characteristics of the users. In some call-handling applications, the caller will encounter the system unexpectedly and infrequently, in others they will have made a deliberate decision to access a value-added service or may be regular users.

Typically, TouchTone dialogues are based upon a menu structure, with dialogue prompts identifying options available. Complex services with a large number of options may require a hierarchical menu structure. The style guidelines discussed provide recommendations on issues such as the maximum size of individual menus and the order in which entries should be placed. A key requirement is that it is clear to the user both where in the dialogue they are and how to navigate around it. Usually the user can interrupt or pre-empt an outgoing menu.

There is a temptation when developing speech dialogues in domains for which TouchTone dialogues have already been developed to simply duplicate the menu structure and produce a 'speech enabled dialogue'. However, being able to select a recognition vocabulary pertinent to the application reduces the need to constantly explain the mapping between the user interface and system functions and can allow the development of more natural 'speech-centric' dialogue structures.

The majority of speech applications currently under development or deployed are 'system directed' and use finite state-based controllers. Here the system 'steers' the caller by imposing a clearly structured dialogue in which the current focus of the interaction is always clearly under the control of the system, and ensures that the caller uses an appropriate vocabulary and a form of speech which is compatible with the recognition technology used. For example, the dialogue may explicitly use prompts such as:

> ***'Speaking after the tone, please say the name of the service which you require: balance, transfer or cheque-book <tone>'***

Such a message tells the user both when they can speak and what the valid responses are. However, such messages can become tiresome and in services where the caller has to navigate through a number of such prompts, a simpler message may initially be offered falling back to more direct phrasing only when necessary.

Typically, a dialogue would have to know how to respond to cases in which the user remains silent or an unexpected response is encountered. For example, a segment of dialogue might be:

> ***Which service do you require?***
> <Silence>
> ***I'm sorry I missed that. Please say the name of service which you require.***
> Help
> ***The following three services are available, balance, transfer and checkbook. Which service do you require?***
> Balance

Although 'system-directed' dialogues have proved effective in applications using isolated-word speech recognition, they are less appropriate in the case of fluent speech entry. Here users are more likely to use complex linguistic forms and will tend to directly steer the dialogue. In these 'mixed initiative' dialogues, the system must keep track of the purpose of the conversation as well as the current active focus, allowing the user more flexibility in taking control of the interaction. In the extreme, the system takes on the role of an assistant, prompting for clarification and confirmation when required. For example [10]:

> ***Which train do you require?***
> I want to go to London this morning

In order to allow such interactions, a more general model of dialogue is required which deals with high-level dialogue processes such as goal seeking, focus shifting and confirmation. The dialogue would typically be expressed in terms of this model, rather than the individual states of an application-specific finite-state dialogue [11].

Alternatively, where a large number of example user interactions with a particular automated application are available, it has been shown experimentally that a statistical model of the dialogue can be generated. This offers the potential for an interesting unification of representations for speech understanding from low-level signal processing to high-level pragmatics.

Special consideration needs to be given in the case of 'computer mediated dialogues' where the automatic system acts as an agent between two people, for example, in the reverse charge system described below or in simultaneous translation systems.

Although much work continues worldwide, arguably no adequate general model of human/machine and machine-mediated speech dialogue has yet been developed.

3.3.2 Dialogue implementation

For even relatively straightforward dialogues, a dialogue control module can become complex, particularly if application and data manipulation functions are not separated from the main logic. As a result, much work has been done in the area of language and tool support for speech dialogues. The aim of such activities is to define a suitable representation scheme for the dialogue which allows user interface design to be separated from other system functions.

Tool support can significantly simplify the development process, reduce the time taken to develop new applications and encourage the use of best practice. The BT VISAGE service creation tool [12, 13] is a portable graphical design environment for speech platforms including the BT ISAP. Many of the applications described in this paper were implemented using VISAGE.

3.3.3 Dialogue design cycle

Developing an effective dialogue is not a straight-forward process and an iterative approach is normally required, taking advantage of service creation tools such as VISAGE and rapid vocabulary creation facilities.

Early prototypes may be evaluated in small scale experiments with potential users, before proper system field trials take place.

In some cases where the requirements of the final system are not well understood or if a suitably accurate prototype of the recognition subsystem is not available, 'Wizard-of-Oz' experiments may be used in which the place of the recogniser is taken by a trained operator or 'wizard'.

The aim of this approach is to allow users to interact with an accurate simulation of the system. Varying portions of the system may be simulated, and appropriate constraints applied to the ways in which the 'wizard' can behave. For example, if the wizard is used solely to simulate the speech recogniser, it will only be able to hear and interpret the users' responses and enter that interpretation into the system, it will not be able to respond directly. As the recogniser will have a restricted vocabulary or grammar, the system may restrict the valid inputs which can be accepted. The sequencing of the dialogue and the selection and output of messages would typically be the responsibility of the system.

Care must be taken to ensure that any aspects in which the simulation may differ from the final system, such as the level of robustness to unexpected responses, are taken into account in the interpretation of results.

Usually, the design of a dialogue will be adapted during trials and after deployment to take into account the observed behaviour and responses of callers and their demographics. This process may include the modification of messages which are causing unexpected problems for callers, the extension of the recognition vocabularies or the statistical weighting of some words within them. Also, as described above the speech recognisers can be optimised by using task-specific data.

In this respect, the dialogue design cycle can be viewed as extending beyond the initial deployment.

4. Approaches to automation

There are two primary approaches to the introduction of speech technology into new or existing applications, and each places different emphases on the underlying speech technology, service creation and effective dialogue design.

- Full automation

  A discrete portion of the service or business is identified and fully automated on a unique access number. Handover to a help-desk or operator may be supported, but this is secondary to the main mode of customer interaction. This is often an appropriate approach to the development of a new service, or the repackaging of an existing service (for example, for market expansion or re-pricing). In both of these cases, the application can be explicitly positioned.

- Progressive automation

  A framework is developed in which speech technology and operators are used co-operatively, the aim being to enable skilled staff to be deployed and operate as effectively as possible.

In the simplest case, an 'auto-attendant' may be responsible for routeing incoming calls to the correct operator, individual, business system or other automated system. Alternatively, customers may be engaged in some preliminary dialogue phase, such as customer identification and verification, before being passed on to the operator.

Alternatively, those calls suitable for full or partial automation may be filtered automatically, according to the user's response to an initial prompt or dialogue. Partial vocabulary recognisers (PVR), as described above, can be useful in applications of this sort. For example, if a user is to be prompted to provide a city name, the PVR may be trained to detect that subset of expected responses which identifies calls suitable for further automation. Those calls falling outside the subset can be routed directly to the operator and the initial response played. In this case, an automated dialogue may be used to hide the playback process. This allows the maximum proportion of calls to be automated given the capabilities of the available recognition components.

Whittaker et al [14] discuss approaches to the use of IVR within call centres.

5. Issues in speech application design

In the design of any speech application a number of issues must be taken into account, for example the role of user priming material and consistency among related applications. In the case of complex automation applications a number of additional factors should be considered.

5.1 *User knowledge*

The amount of domain information known by the caller may be limited and inaccurate. The designer may wish to concentrate on information which is likely to be correctly known by the caller, depending on less reliable information only when necessary or during confirmation. In addition, some information may be intrinsically or culturally variable such as job titles and business functions.

5.2 *Normalisation of application databases*

Many application databases are extremely irregular, with multiple spellings, spelling errors, variable punctuation and inconsistent abbreviations. Some normalisation may be required.

5.3 *Confidence and confirmation*

In many applications the caller is asked for a series of pieces of information which are then used by the agent to conduct some database search. Often this will result in the operator being presented with several screens of information which may be intelligently browsed. Replacing this behaviour within a speech system is a complex process, which may typically result in more complex confirmation strategies and the acquisition of additional pieces of information.

5.4 Operator or agent interworking

The role of the operator or agent within an automated system is very important. The design of the system, and hence the dialogue and style of the application, may be biased to emphasise either or both. For example, the operator may be used as a fall-back 'help-desk', in a 'pop-up' mode monitoring the progress of multiple calls, or as the prime focus of the application with the automatic system portrayed as a distinct phase. This leads to a spectrum from 'stealthy' automation in which the caller is unaware that the call is being fully or partially automated to more explicit systems which may be deliberately styled to emphasise the point, perhaps by some handover signal, such as ring tone or a change of voice.

5.5 Vocabulary processing

Particular care must be taken in the use of certain vocabularies. Name information requires additional modelling of homophones (words which sound the same but have different spellings, e.g. Miller and Millar) and synonyms (alternative words with the same meaning, e.g. Jim and James). Typically 30% of names may be homophones or synonyms of others. Place names may require complex locality modelling, as separate users may use different and overlapping terms for the same places.

5.6 Partial versus full spelling

Whereas it may be reasonable to expect a caller to spell their own surname, it is less likely that a caller entering the spelling of another individual's name will do so successfully. Partial spelling of the first N letters may be more appropriate, being less error prone, less variable and less time consuming. Partial spelling raises issues as to performance, resolving power and human factors.

6. Speech platforms

Speech applications vary significantly in scale and complexity, from single line systems residing in a home PC, through small office-based servers with some tens of lines and call-centre systems with perhaps hundreds of lines, to embedded network systems capable of supporting many hundreds or thousands of simultaneous calls.

In order to support this variety of services a range of platforms have become available, both at the periphery of the network and within it.

Large switches and some PABX products offer restricted TouchTone facilities. IVR platforms aim to provide a scalable, reusable and flexible component tailored to the delivery of multichannel voice solutions.

Such platforms usually provide the core components of TouchTone, speech input, output and storage and a graphical service creation environment. In addition, they support facilities for integration with application host computers and telco functions such as billing, alarm handling and service management. The BT integrated speech applications platform (ISAP) [5, 6] is such a platform now deployed within the BT network. Many of the applications discussed below are implemented on the ISAP.

Standards are emerging for use in the integration of call and speech-handling functions with desktop and call-centre applications — for example Dialogics SCSA, Microsoft's TAPI and SAPI, and TSAPI from Novell and the Versit consortium. Along with the availability of low-cost speech technology and customer premises platforms, these will encourage the growth in the use of speech technology within applications at the periphery of the network as well as within it.

In many cases a designer has the choice between a customer premises-based solution and one based on a managed service provided by the network or other service provider. Whittaker et al [14] discuss some of these issues with relevence to the use of IVR within call centres.

7. Example voice applications

This section describes a number of current and emerging voice services.

7.1 Voice messaging

Voice-mail systems have been widespread within the corporate environment for some years now. Typically, systems have been operated either as a corporate resource or as a managed service provided by the network or operator or service provider. They normally support a complex structured TouchTone dialogue and are aimed primarily at the business user.

Speech recognition offers the potential both to simplify the user interface and to make the functionality available to a wider audience. BT's CallMinder ™ service [15] provides a mass-market residential network-based call-answering system using speaker-independent speech recognition.

This is an example of how a major new service can be introduced using speech technology. Through the use of an easy-to-follow directed dialogue and a relatively small vocabulary, the system can be used by any user without the need for a complex instruction card or TouchTone phone, although TouchTone can be used according to user preference.

7.2 Call delivery

Uncompleted calls, either through number busy or no answer, are a cause of frustration to callers and represent a loss of revenue to the operator. Systems such as CallMinder tackle this problem from the perspective of the called party; message delivery systems approach it from the other direction by offering the caller the ability to leave a message for later delivery or distribution. Typically the system may attempt to deliver the message a number of times over a set period or at a particular time.

Payphone voice messaging is an example of this and provides a delivery facility from payphones. A system has been successfully trialled in a number of locations, including motorway service stations and major London railway stations, to assess the impact of such a service. The very nature of the service means that all calls are made from pay-phones many of which are sited in high ambient noise locations, providing a specific challenge to speech recognition. In this case, the dialogue was extremely straightforward and was implemented using TouchTone.

This application is currently being developed as a facility for BT's new cashless services platform which is a replacement for the BT Chargecard (calling card) platform.

This is a simple example of how speech technology can begin to convert the telephone network from a seemingly simple system to a more intelligent assistant.

7.3 Operator assistance

Speech technology provides many opportunities for the streamlining of the operator assistance service. Typically, once the nature of many calls has been identified they can be completed automatically, either through a simple message, call redirection or an automated dialogue. Examples include:

- calls to operator assistance which are normally chargeable services provided on another access number,
- calls which require the operator to act as an intermediary in a routine dialogue such as that associated with reverse charge (call collect) calls,
- advice duration and charge (ADC),
- alarm calls.

These applications are described in Orrey et al [16].

7.4 Call centres

Call centres play an increasingly important part in the way many companies manage contacts with their customers. They provide the focal point for telephone contacts both inbound to the business and outbound from it. IVR can play a key role in providing a high quality of service to customers and optimising the performance of the centres. For further detail, see Whittaker et al [14].

8. Next generation systems

This section describes some of the prototype systems which have been developed at BT Laboratories, and which give some indication of the way in which speech technology can be used to approach the development of advanced applications.

8.1 PaymentLine

'PaymentLine' was developed as a prototype system on which to develop technologies for advanced applications such as home shopping and customer handling. It demonstrates some of the generic aspects of automated call centres and illustrates the automation of customer bill payment. Customer identification is a problem common to many applications. For each customer, the database contains:

- an identification number, name and address,
- payment methods used to date,
- financial information relating to the account.

The system aims to:

- fully automate all calls (i.e. no operator involvement in the majority of calls),
- allow the caller to select a new method of payment and make specific payments,
- record 'problem' calls for later off-line transcription.

Good evening. Welcome to BT's PaymentLine. First of all, I need to find out who you are. Please say the last 8 digits of your reference number:
Double-8-3-7-double-3-double-7
After the tone, please give your surname:
Smith
Is your surname Smith?
Yes
Thank you. Please wait while I check your details. The full balance of your account is £43.71
...
Please let me know how much you would like to pay by saying full balance, monthly payment or other amount.
...

8.2 Directory automation

Access to directory information is a key issue for many companies, both corporate directories within companies and directory assistance information in the case of network operators. Speech technology can be used to supplement existing methods for accessing directory information. discusses. The implementation of a corporate directory system is discussed elsewhere [3].

A series of demonstration and trial systems have been developed at BT Laboratories to investigate the feasibility of various approaches to the full and partial automation of directory assistance (DA) [1].

In the UK, DA is a chargeable service with a database of 20M residential business and government entries involving half a million distinct surnames and nearly 30K distinct localities of differing sizes. One such prototype dialogue is shown below and shows the use of large-vocabulary spoken and spelt recognition over the telephone network to access a subset of the database.

Directory Enquiries. Which town please?
Martlesham
Do you know the road name?
Yes
Please say the complete road name
Viking Heights
And now say the surname
Foster
And now spell the surname
F-O-S-T-E-R
Is the surname Foster?
Yes
The number is ...

Due to the complexity of this application and the size of the vocabulary and application database, the automation of directory assistance will remain a significant challenge to speech recognition for some time.

9. Conclusions

This paper has discussed various aspects of the development of interactive speech systems with a particular emphasis on telecommunications applications. In addition a number of example services and advanced prototypes have been described.

Speech technology offers great promise for high quality user interfaces to a wide variety of applications and services. It is rapidly becoming a vital enabling technology for a wide range of business and consumer applications, enabling services to be made available rapidly to large markets.

Although, it is widely agreed that there have been few substantial theoretical breakthroughs in speech recognition technology within the last decade, the art of engineering high-quality speech systems has reached the level at which it now has a major impact on product and systems development.

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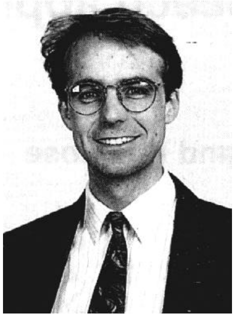

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